



Manual for
Real-Time Quality Control of In-situ Temperature and Salinity Data

A Guide to Quality Control and Quality Assurance for In-situ Temperature and Salinity Observations

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Document Validation



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Revision History

Date	Revision Description	Notes
12/2013	Original Document Published.	
01/2016	Revise cover page to include new IOOS logo. Revise dates on *Document Validation* page and substitute new logo (page ii). Add statement requesting feedback from *Manual Users* (page vi). Update *Acknowledgements* to include Version 2.0 team members (page vii). Update *Acronyms and Abbreviations* (page viii). Update definition of real time in *Definitions of Selected Terms* (page x). Revise *Background and Introduction* to reflect updated temperature/salinity manual, as well as additional manuals that have been developed (page 1). Revise section 2.0 content in various places to reflect feedback from reviewers (pages 3-4). Add section 2.5 with content addressing data uncertainty (page 12). Update content in sections 3.1 and 3.2 (pages 13-14). Update language in section 4.0, *Summary* (page 31). Update *References* and *Supporting Documents* (page 32-35). Update Temperature and Salinity Manual Team members (page B-1).	Manual updated with revisions listed sequentially
03/2020	Revise cover page to include correct date and version number and also to comply with accessibility requirements; update page numbers. Revise dates and signature blocks on *Document Validation* page (page ii). Update the email address to *Manual Users* for submitting information about efforts to implement QARTOD processes (page vi). Update *Acknowledgements* to include Version 2.1 team members (page vii). Update *Acronyms and Abbreviations* (page viii). Add definition of interoperable, sensor, and variable in *Definitions of Selected Terms* (page x). Revise *Background and Introduction* to reflect updated temperature/salinity manual, as well as additional manuals that have been developed (page 1). Revise section 2.0 content in various places to reflect feedback from reviewers (pages 3-5); update table 2-1, update photo in fig. 2-4, caption in fig. 2-5, and caption in fig. 2-7. Update content in sections 3.1, 3.2, and 3.3 (specifically Tests 3, 7, 8, and tables 3-4 and 3-5. Update language in section 4.0, *Summary* (page 27). Update *References* and *Supporting Documents* (page 28–31). Update Temperature and Salinity Manual Team members (page A-1).	Second update with revisions listed sequentially

Endorsement Disclaimer

Mention of a commercial company or product does not constitute an endorsement by NOAA. Use of information from this publication for publicity or advertising purposes concerning proprietary products or the tests of such products is not authorized.

Request to Manual Users

To gauge the success of the QARTOD project, it helps to be aware of groups working to utilize these QC tests. Please notify us of your efforts or intentions to implement QARTOD processes by sending a brief email to qartod.board@noaa.gov or posting a notice at http://www.linkedin.com/groups?gid=2521409.

Acknowledgements

Special thanks go to members of the Temperature/Salinity Manual committee, who contributed their expertise to develop the content of the original manual and also to document reviewers, whose many valuable suggestions greatly enhanced the manual content.

Thanks for assistance with the 2015 update go to Igor Shkvorets (RBR Ltd.) for his review and his expertise regarding standard seawater and instrument calibration. We appreciate the additional photographs and thoughtful review provided by Stephanie Jaeger at Sea-Bird Electronics.

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We also appreciate the efforts of Stephanie Smith and Inger M. Graves (Xylem/Aanderaa) to obtain the photo of the 4419R sensor. And finally, we thank Annie Wong (University of Washington) for her very helpful thoughts regarding comparisons of Argo QC tests to QARTOD tests.

Acronyms and Abbreviations

AOML	(NOAA) Atlantic Oceanographic and Meteorological Laboratory
AOOS	Alaska Ocean Observing System
AUV	Autonomous Underwater Vehicle
CDIP	Coastal Data Information Program
CeNCOOS	Central and Northern California Ocean Observing System
CO-OPS	(NOAA) Center for Operational Oceanographic Products and Services
CRC	Cyclic Redundancy Check
CTD	Conductivity, Temperature, and Depth
EuroGOOS	European Global Ocean Observing System
GCOOS	Gulf of Mexico Coastal Ocean Observing System
GLOS	Great Lakes Observing System
GOOS	Global Ocean Observing System
GPS	Global Positioning System
GTSPP	Global Temperature-Salinity Profile Program
IAPSO	International Association for the Physical Sciences of the Oceans
IMOS	Integrated Marine Observing System
IOC	Intergovernmental Oceanographic Commission
IODE	International Oceanographic Data and Information Exchange
IOOS	(U.S.) Integrated Ocean Observing System
MARACOOS	Mid-Atlantic Regional Association Coastal Ocean Observing System
NANOOS	Northwest Association of Networked Ocean Observing Systems
NCEI	(NOAA) National Centers for Environmental Information (formerly NODC)
NDBC	(NOAA) National Data Buoy Center
NERACOOS	Northeastern Regional Association of Coastal Ocean Observing Systems
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NODC	(NOAA) National Oceanographic Data Center (now NCEI)
P	Pressure
PacIOOS	Pacific Islands Ocean Observing System
PMEL	Pacific Marine Environmental Laboratory
PSS-78	Practical Salinity Scale-1978
QARTOD	Quality-Assurance/Quality Control of Real-Time Oceanographic Data
QA	Quality Assurance

QC	Quality Control
RCOOS	Regional Coastal Ocean Observing System
SCCOOS	Southern California Coastal Ocean Observing System
SD	Standard Deviation
SECOORA	Southeast Coastal Ocean Observing Regional Association
SIO	Scripps Institution of Oceanography
SP	Practical Salinity
TS	Temperature/Salinity
UNESCO	United Nations Educational, Scientific and Cultural Organization
USGS	United States Geological Survey

Definitions of Selected Terms

This manual contains several terms whose meanings are critical to those using the manual. These terms are included in the following table to ensure that the meanings are clearly defined.

Codable Instructions	Codable instructions are specific guidance that can be used by a software programmer to design, construct, and implement a test. These instructions also include examples with sample thresholds.
Data Record	A data record is one or more messages that form a coherent, logical, and complete observation.
Interoperable	Interoperable means the ability of two or more systems to exchange and mutually use data, metadata, information, or system parameters using established protocols or standards.
Message	A message is a standalone data transmission. A data record can be composed of multiple messages.
Operator	Operators are individuals or entities who are responsible for collecting and providing data.
Practical Salinity (SP)	A unitless ratio expressing salinity as defined by the Practical Salinity Scale 1978 (PSS-78).
Quality Assurance (QA)	QA involves processes that are employed with hardware to support the generation of high quality data. (section 2.0 and Bushnell et al. 2019)
Quality Control (QC)	QC involves follow-on steps that support the delivery of high quality data and requires both automation and human intervention. (section 3.0)
Real-Time	Real-time means that data are delivered without delay for immediate use. The time series extends only backwards in time, where the next data point is not available. Data update latency can range from a few seconds to a few hours or even days, depending upon the variable. (section 2.0)
Sensor	A sensor is a device that detects or measures a physical property and provides the result without delay. A sensor is an element of a measuring system that is directly affected by a phenomenon, body, or substance carrying a quantity to be measured. (JCGM 2012)
Threshold	Thresholds are limits that are defined by the operator.
Variable	A variable is an observation (or measurement) of biogeochemical properties within oceanographic and/or meteorological environments.

1.0 Background and Introduction

The U.S. Integrated Ocean Observing System (IOOS®) has a vested interest in collecting high-quality data for the 34 core variables (https://ioos.noaa.gov/about/ioos-by-the-numbers) measured on a national scale. In response to this interest, U.S. IOOS continues to establish written, authoritative procedures for the quality control (QC) of real-time data through the Quality Assurance/Quality Control of Real-Time Oceanographic Data (QARTOD) project, addressing each variable as funding permits. This manual on the real-time QC of temperature and salinity data was first published in December 2013 as the fourth core variable to be addressed and was updated in December 2015; this is the second update. Other QARTOD guidance documents that have been published by the U.S. IOOS project to date are listed below and are available at https://ioos.noaa.gov/project/qartod/. They are also available from the NOAA repository at https://repository.library.noaa.gov and at the IOC / GOOS Ocean Best Practice System at https://www.oceanbestpractices.org/. Users are encouraged to check the IOOS website to ensure they have the most recent version of the manuals.

1) U.S. Integrated Ocean Observing System, 2017. U.S IOOS QARTOD Project Plan - Accomplishments for 2012–2016 and Update for 2017–2021. 48 pp. https://doi.org/10.7289/V5JQ0Z71.

2) U.S. Integrated Ocean Observing System, 2018. Manual for Real-Time Quality Control of Dissolved Oxygen Observations Version 2.1: A Guide to Quality Control and Quality Assurance for Dissolved Oxygen Observations in Coastal Oceans. 53 pp. https://doi.org/10.25923/q0m1-d488

3) U.S. Integrated Ocean Observing System, 2019. Manual for Real-Time Quality Control of In- Situ Surface Wave Data Version 2.1: A Guide to Quality Control and Quality Assurance of In- Situ Surface Wave Observations. 69 pp. https://doi.org/10.25923/7yc5-vs69

4) U.S. Integrated Ocean Observing System, 2019. Manual for Real-Time Quality Control of In-Situ Current Observations Version 2.1 A Guide to Quality Control and Quality Assurance of Acoustic Doppler Current Profiler Observations. 54 pp. https://doi.org/10.25923/sqe9-e310

5) U.S. Integrated Ocean Observing System, 2016. Manual for Real-Time Quality Control of Water Level Data Version 2.0: A Guide to Quality Control and Quality Assurance of Water Level Observations. 46 pp. https://doi.org/10.7289/V5QC01Q7

6) U.S. Integrated Ocean Observing System, 2017. Manual for Real-Time Quality Control of Wind Data Version 1.1: A Guide to Quality Control and Quality Assurance of Coastal and Oceanic Wind Observations. 47 pp. https://doi.org/10.7289/V5FX77NH.

7) U.S. Integrated Ocean Observing System, 2017. Manual for Real-Time Quality Control of Ocean Optics Data Version 1.1: A Guide to Quality Control and Quality Assurance of Coastal and Oceanic Optics Observations. 49 pp. https://doi.org/10.25923/v9p8-ft24.

8) U.S. Integrated Ocean Observing System, 2018. Manual for Real-Time Quality Control of Dissolved Nutrients Data Version 1.1: A Guide to Quality Control and Quality Assurance of Coastal and Dissolved Nutrients Observations. 56 pp. https://doi.org/10.7289/V5TT4P7R

9) U.S. Integrated Ocean Observing System, 2016. Manual for Real-Time Quality Control of High Frequency Radar Surface Currents Data Version 1.0: A Guide to Quality Control and Quality Assurance of High Frequency Radar Surface Currents Data Observations. 60 pp. https://doi.org/10.7289/V5T43R96

10) U.S. Integrated Ocean Observing System, 2017. Manual for Real-Time Quality Control of Phytoplankton Data Version 1.0: A Guide to Quality Control and Quality Assurance of Phytoplankton Data Observations. 67 pp. https://doi.org/10.7289/V56D5R6S

11) U.S. Integrated Ocean Observing System, 2017. Manual for Real-Time Quality Control of Passive Acoustics Data Version 1.0: A Guide to Quality Control and Quality Assurance of Passive Acoustics Observations. 43 pp. https://doi.org/10.7289/V5PC30M9

12) U.S. Integrated Ocean Observing System, 2018. Manual for Real-Time Quality Control of Stream Flow Data Version 1.0: A Guide to Quality Control and Quality Assurance of Stream Flow Observations in Rivers and Streams. 46 pp. https://doi.org/10.25923/gszc-ha43

13) U.S. Integrated Ocean Observing System, 2019. Manual for Real-Time Quality Control of pH Data Version 1.0: A Guide to Quality Control and Quality Assurance of pH Data Observations. 56 pp. https://doi.org/10.25923/111k-br08

Please reference this document as:

U.S. Integrated Ocean Observing System, 2020. Manual for Real-Time Quality Control of In-situ Temperature and Salinity Data Version 2.1: A Guide to Quality Control and Quality Assurance of In-situ Temperature and Salinity Observations. 50 pp. https://doi.org/10.25923/x02m-m555

This manual is a living document that reflects the state-of-the-art QC testing procedures for temperature and salinity observations. It is written for the experienced operator but also provides examples for those who are just entering the field.

2.0 Purpose/Constraints/Applications

This manual documents a series of test procedures for quality control (QC) of temperature and salinity (TS) data. TS observations covered by these procedures are collected in oceans, coastal waters, and lakes in real-time. The scope of real-time has expanded to accommodate the span of the 34 variables covered by U.S. IOOS. The characteristics of real-time (in no particular order) are:

- data delivered as soon as possible after acquisition for immediate use
- a time series extending only backwards in time, where the next data point is not available
- sample intervals from a few seconds to a few hours or even days, depending upon the sensor configuration

The tests draw from existing expertise in programs such as the Global Temperature and Salinity Profile Programme (GTSPP) and Argo (Wong et al, 2020). The Global Climate Observing System (GCOS) recognizes the GTSPP as one of the international operational activities that provide essential, sub-surface climate variables of temperature and salinity profile data. GTSPP provides timely and complete data with documented quality flags and implements internationally accepted quality control and overall management of ocean data fully in accordance with the GCOS action plan (www.nodc.noaa.gov/GTSPP/). The Argo program is a global array of 3,000 free-drifting profiling floats measuring the temperature and salinity of the upper 2,000 meters (m) of the ocean. The program provides continuous monitoring of the temperature, salinity, and velocity of the upper ocean, with all data being relayed and made publicly available within hours after collection (www.argo.net).

This manual differs from existing QC procedures for TS data in that its focus is on real-time, and it is not constrained to deep oceans, as are GTSPP and Argo. It presents practices and procedures from these programs as a basis for developing codable instructions and provides guidance for the broader ocean observing community. These existing programs and others within the observing community use many of the same sensors. The tests and codable instructions described herein are examples that might be employed. But, operators may choose to use similar tests from existing programs (such as the MATLAB®-coded QC tests posted by the Integrated Marine Observing System [IMOS] at https://github.com/aodn/imos-toolbox) or to develop their own tests to accomplish the same results.

High-quality marine observations require sustained quality assurance (QA) and QC practices to ensure credibility and value to operators and data users. Some QA practices involve processes that are employed with hardware to support the generation of high-quality data, such as a sufficiently accurate, precise, and reliable sensor with adequate resolution. Others include sensor calibration; calibration checks and/or in-situ verification, including post-deployment calibration; proper deployment considerations, such as measures for corrosion control and anti-fouling; reliable data communications; adequate maintenance intervals; and creation of a robust QC process. Post-deployment calibration (instrument verification after recovery) issues are not part of the scope of this manual. Although QC and QA are interrelated and important to the process, QA issues are only briefly addressed here. Bushnell et al. (2019) provide a more comprehensive review of QA processes, including several example checklists in the Supplementary Material.

QC involves follow-on steps that support the delivery of high-quality data and requires both automation and human intervention. QC practices include such things as data integrity checks (format, checksum, timely arrival of data), data value checks (threshold checks, minimum/maximum rate of change), neighbor checks, climatology checks, model comparisons, signal/noise ratios, the mark-up of the data, the verification of user satisfaction, and generation of data flags (Bushnell 2005).

These procedures are written as a high-level narrative from which a computer programmer can develop code to execute specific data flags (data quality indicators) within an automated software program. A code repository exists at https://github.com/ioos/ioos_qc where operators may find or post examples of code in use. Although certain tests are recommended, thresholds can vary among operators. The tests described here are designed to support a range of TS

sensors and operator capabilities. Some well-established programs with the highest standards have implemented very rigorous QC processes. Others, with different requirements, may utilize sensors with data streams that cannot support as many QC checks—all have value when used prudently. Users must understand and appropriately utilize data of varying quality, and operators must provide support by documenting and publishing their QC processes. A balance must be struck between the time-sensitive needs of real-time observing systems and the degree of rigor that has been applied to non-real-time systems by operators with decades of QC experience.

These tests apply only to the in-situ, real-time measurement of TS as observed by sensors deployed on rigidly mounted, moored, or moving platforms (e.g., drifting buoys, autonomous marine vehicles, ships) but not to remotely sensed TS measurements (e.g., satellite observations).

The process of ensuring data quality is not always straightforward. QA/QC procedures may be specific to a sensor technology or even to a particular manufacturer's model, so the establishment of a methodology that is applicable to every sensor remains challenging.

Sensors deployed on mobile platforms such as gliders require attention to proper QA procedures both before and after the deployment (see Bushnell et al. [2019] for general QA guidance). While outside the scope of the real-time tests described in this manual, the manufacturer's recommendations for factory calibration schedules and proper sensor maintenance must be followed.

The instruments described in figs. 2-1 through 2-6 are illustrations provided by manufacturers and TS committee members. They may be referred to as TS (temperature and salinity), CTD (conductivity, temperature and depth) or CT sensors (conductivity and temperature), and they directly measure T, C, and P (pressure). Their measurements are used to derive salinity, depth, density, specific gravity, and specific conductance. Table 2-1 lists companies that produce sensors covered in this manual.

Table 2-1. TS sensor manufacturers

Aanderaa
Campbell Scientific
Greenspan
Hach
In-Situ
JFE Advantech Company Ltd.
RBR Ltd.
Rockland Scientific International Inc.
Sea-Bird Scientific
YSI

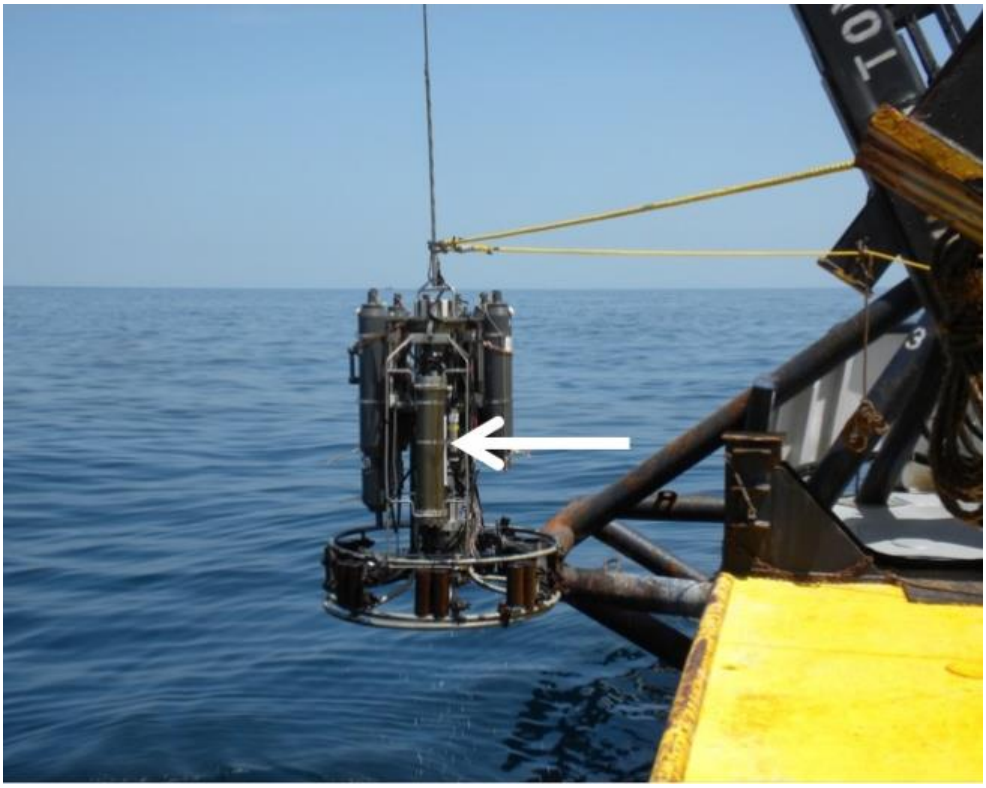


Figure 2-1. A profiling Sea-Bird Scientific SBE 9plus CTD mounted on a rosette with Niskin bottles is recovered during a cruise aboard the R/V *Ocean Veritas* following the 2010 Deepwater Horizon incident (photo courtesy of Mark Bushnell).

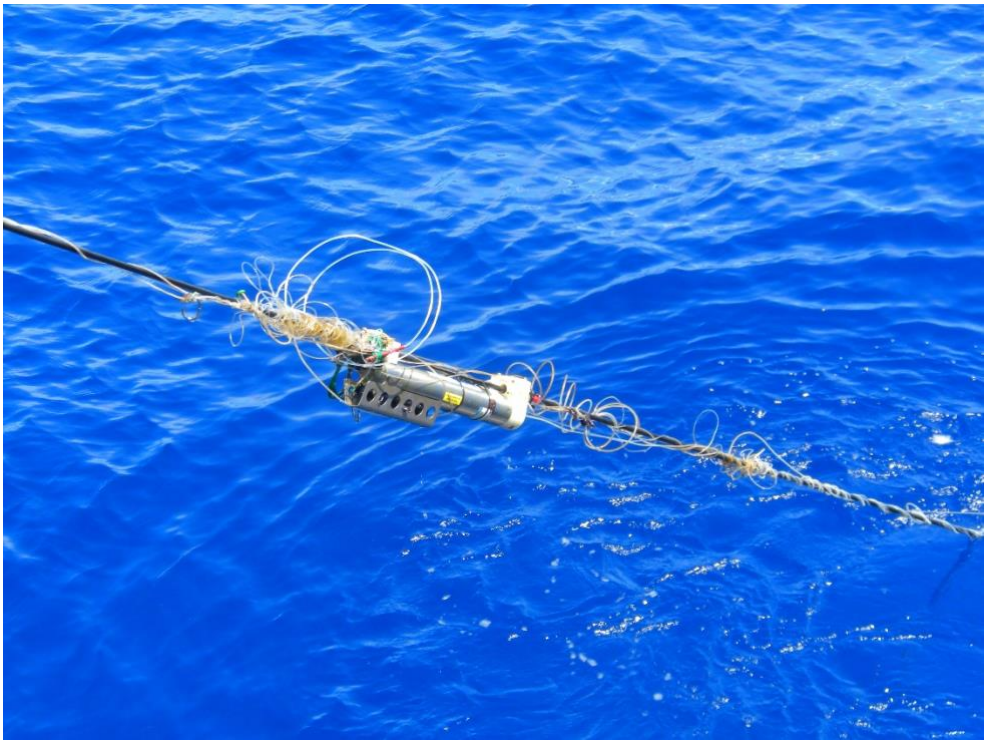


Figure 2-2. This Sea-Bird Scientific SBE 37-IM temperature and conductivity recorder uses an inductive modem to transmit data up the mooring cable to a surface receiver. This sensor has become entangled in fishing line (photo courtesy of Rick Cole/RDSEA International, Inc.).



Figure 2-3. This towed RBR concerto CTD uses an inductive conductivity sensor with an external field and no pump (photo courtesy of Igor Shkvorets/RBR Ltd.).

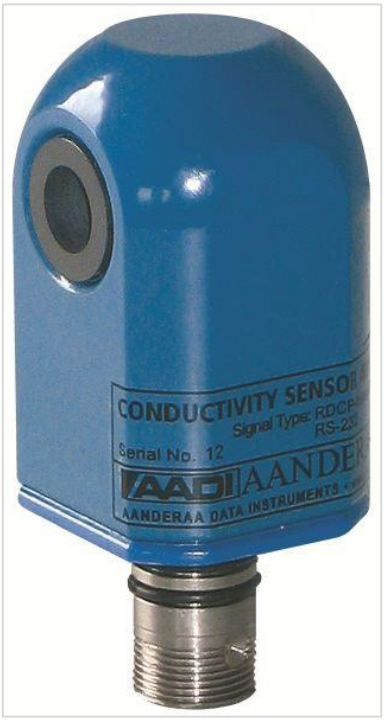


Figure 2-4. The Aanderaa/Xylem 4419R is an example of an inductive sensor with an external field. Operators must be certain that additional hardware is sufficiently distant from the toroid to avoid interference. This sensor is designed for simple and easy cleaning (photo courtesy of Paul Devine/Teledyne RD Instruments).



Figure 2-5. The Teledyne RD Instruments Citadel CTD-NH (discontinued in 2019) is an example of an inductive sensor with a constrained field (photo courtesy of Paul Devine/Teledyne RD Instruments).



Figure 2-6. The JFE Advantech INFINITY-CT A7CT-USB (photo courtesy of Fabian Wolk, Ph.D./Rockland Scientific International Inc.).

2.1 Data Processing Methodology

The type of sensor system collecting the data and the system processing and transmitting the measurements can affect which QC algorithms are used. In-situ systems with sufficient onboard processing power within the sensor may process the original (raw) data and produce derived products, such as salinity, density, or speed of sound. If ample bandwidth is available, the entire original data stream may be transmitted ashore and subsequently quality controlled. If lacking sufficient bandwidth, the operator may not be able to apply tests designed for raw data. Therefore, because operators have different data processing methodologies, three levels of QC are proposed: required, strongly recommended, and suggested.

2.2 Traceability to Accepted Standards

To ensure that TS sensors are producing accurate data, rigorous calibrations and calibration checks must be performed in addition to QC checks. Most operators rely upon manufacturer calibrations and conduct calibration checks only before deployment. These calibration checks are critical to ensuring that the manufacturer calibration is still valid. These procedures are currently considered QA and addressed further in Bushnell et al. (2019).

Calibrations and calibration checks must be traceable to accepted standards. The National Institute of Standards and Technology (NIST) (http://www.nist.gov/index.html), a provider of internationally accepted standards, is often the source for these standards. Calibration activities must be tailored to match data use and resources; calibration cost and effort increase dramatically as accuracy requirements increase. NIST standards for temperature and pressure sensors can be met using transfer references such as platinum resistance thermometers and deadweight testers. Conductivity sensors are most commonly calibrated against the International Association of Physical Sciences of the Ocean (IAPSO) standard seawater, certified by Ocean Scientific International Ltd. (OSIL) in terms of the ratio K_{15}. The ocean observing community uses dimensionless practical salinity as defined by the Practical Salinity Scale-1978 (PSS-78), developed in 1978 (UNESCO 1981). PSS-78 is based on an equation relating salinity to the ratio K_{15} of the electrical conductivity of seawater at 15 °C to that of a standard potassium chloride solution (KCl) (http://salinometry.com/pss-78). Laboratory

salinometers (http://salinometry.com/modern-oceanographic-salinometers) are used for the precise measurement of salinity samples during laboratory conductivity calibrations and bottle samples at sea. A new absolute salinity scale was adopted in 2009 by the Scientific Committee on Oceanic Research and the IAPSO Working Group 127 (WG127) (McDougall et al. 2009). However, WG127 has advised the continued use of the PSS-78 for data repositories.

2.3 Sensor Deployment Considerations

TS sensors can be deployed in several ways. Stationary sensor deployments are on fixed platforms or moorings where there is minimal horizontal or vertical movement. Mobile platforms are available in a variety of configurations and require different real-time TS QC considerations. Mobile platforms are, in order of increasing complexity: fixed vertical profilers, mobile surface vessels, and vessels freely operating in three dimensions (e.g., gliders, floats, powered automated underwater vehicles or AUVs). Figures 2-7 through 2-9 illustrate examples.

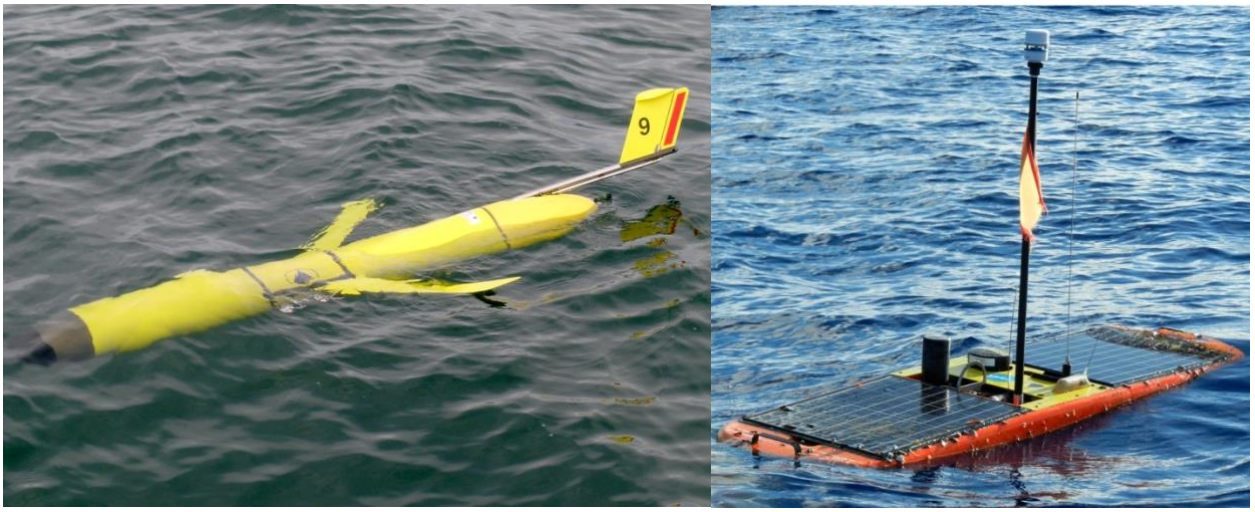


Figure 2-7. Slocum Glider Profiler 3-D (L) and Liquid Robotics Wave Glider® (R) (photos courtesy of Dave Fratantoni, Ph.D.)

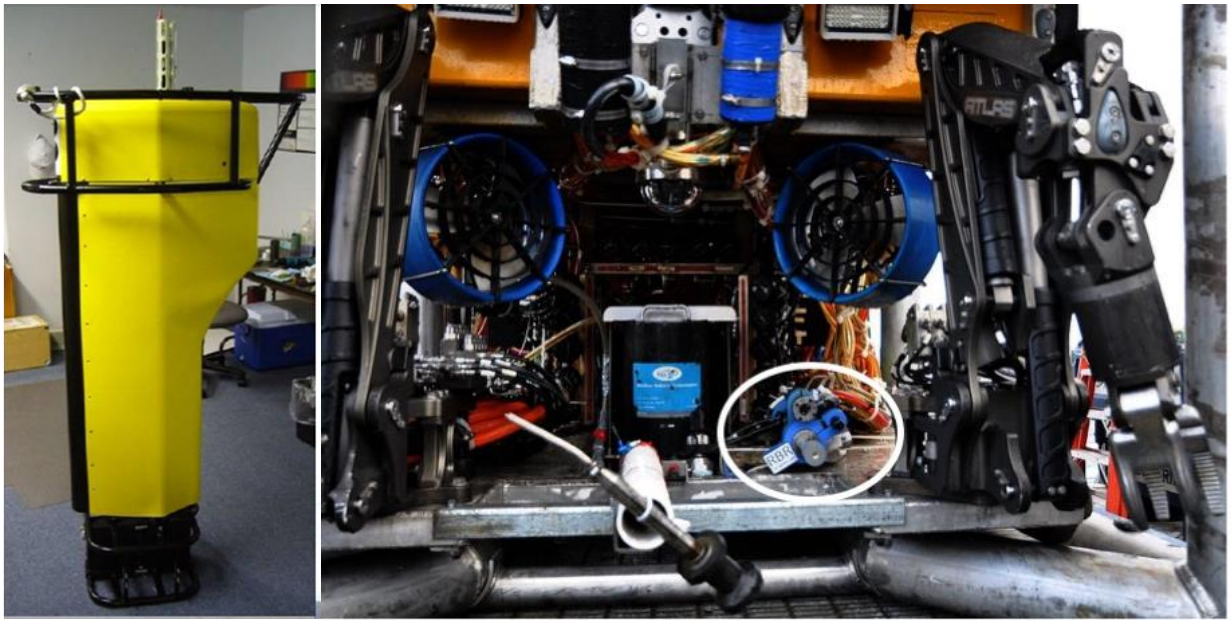


Figure 2-8. WET Labs AMP C100 In-Situ Profiler (courtesy of WET Labs) (L); RBR CTD sensor on an Oceaneering ROV (R) (photo courtesy of Igor Shkovorets/RBR Ltd.).

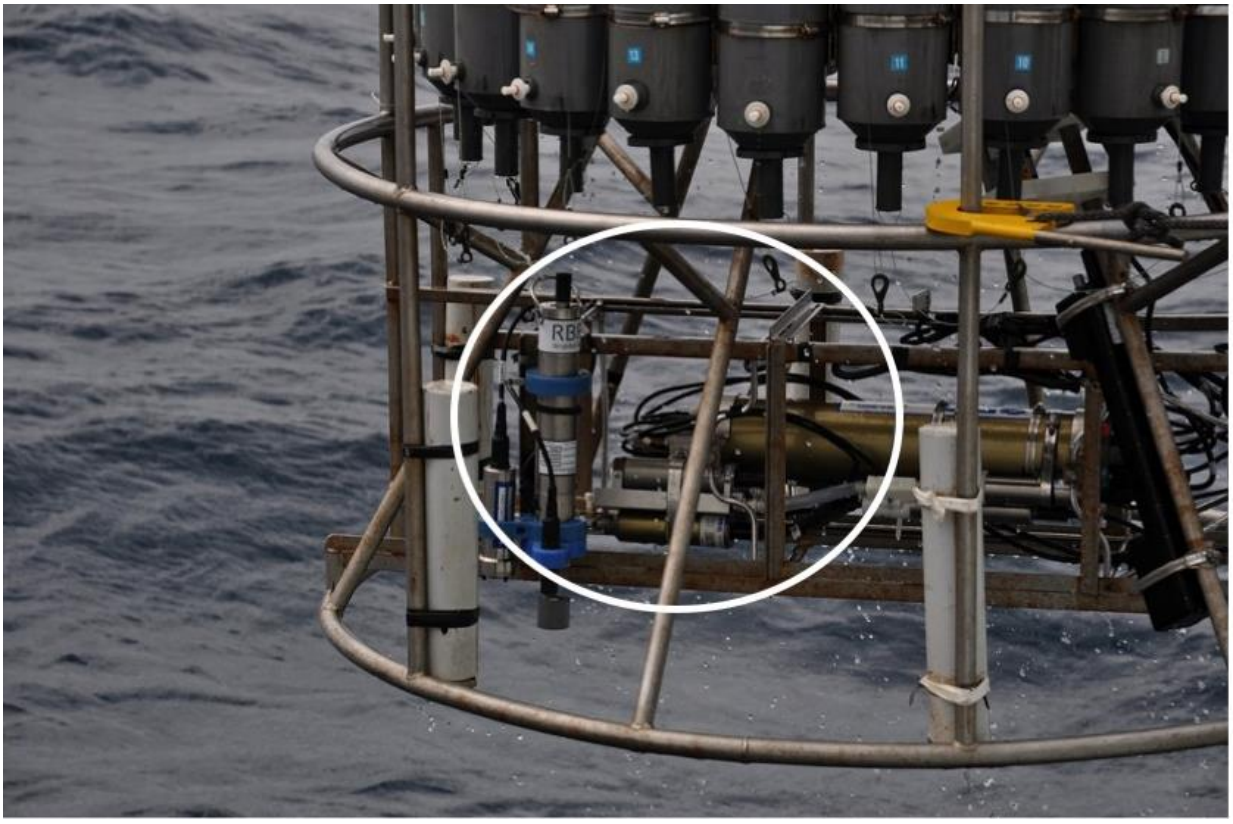


Figure 2-9. This CTD/bottle rosette shows the use of both Sea-Bird Scientific (SBE 9plus) and RBR sensors (photo courtesy of Igor Shkvorets/RBR Ltd.).

Moving Platform Corrections

Mobile and profiling sensors commonly move through gradients over short time scales and require additional QC. Therefore, two additional corrections specifically for mobile and profiling sensors should be applied prior to the real-time QC tests described in this manual: a response time correction and a thermal mass correction. The methods employed to make these corrections are usually developed and provided by the manufacturer, since they are unique to each specific sensor and may even require calibration factors. The following discussion is an overview of the complexity associated with obtaining CTD data of high accuracy but is not meant to instruct or guide operators on these correction processes.

Response Time Correction. The first correction is made because the CT sensors on the instrument have different measurement response times and may have different physical locations; thus, the two independent measurements should be aligned with respect to time so that each CTD record represents a measurement on a single parcel of water. This time shift should account for the sample rate of the instrument and for the known constant flow rate of the pump on the CTD sensor (if pumped) or the speed of the glider through the water column (if unpumped) (Garau et al. 2011).

Thermal Mass Correction. A second correction is needed to account for the thermal mass of the conductivity cell and its effect on the resulting salinity calculation. The CTD sensor temperature is measured outside the conductivity cell, while the conductivity is measured inside the cell. In addition, the conductivity cell can store heat from the surrounding water inside the wall of the cell, resulting in a heating or cooling of new water parcels as they pass through the cell. As a result of this thermal lag, without the corrections, the paired conductivity and temperature used to calculate salinity could result in erroneous salinity values, especially across temperature gradients. A method to correct for heating inside the cell has been developed, resulting in more accurate salinity profiles (Morison et al. 1994). Garau et al. (2011) specifically address the additional considerations associated with unpumped CTD sensors deployed on gliders.

2.3.1 Fixed Platform and Fixed Vertical Profilers

Fixed vertical TS profiles are obtained from a variety of systems, including rigid-mounted systems, buoy/mooring climbers, surface- or bottom-tethered systems, or even casts from regularly scheduled manual station observations. Tests described for a fixed sensor (section 3.3) either remain unchanged or are conducted along the vertical ('z') axis, as well as along a time series of observations.

2.3.2 Mobile Surface Vessels

Examples of mobile surface vessels include manned vessels of opportunity and autonomously operated vehicles, such as wave gliders, fitted with TS sensors. Samples are obtained at a fixed depth along a track and may be taken at fixed temporal or spatial intervals. Tests may be conducted along the vessel path ('s'), or the path may be projected along 'x' (longitude) and 'y' (latitude) coordinates, as well as along a time series of observations.

2.3.3 3-D Profiler Vessels

Sensors mounted on gliders, floats, powered AUVs, and animals can provide TS observations in a wide variety of space/time configurations. Observations can be as simple as along path 's', periodic vertical ascent profiles recorded following at-depth drifts (Argo profilers), or real-time processed down/up profiles (gliders). When applying increasingly complex real-time QC tests to increasingly complex deployments, challenges may arise. However, most of the 13 tests described in section 3.3 can be applied with little modification.

2.4 Hardware Limitations

Most temperature and pressure sensors can withstand moderate bio-fouling. However, conductivity sensors cannot, so salinity observational accuracy gradually degrades with time. Because the performance decline is gradual or can occur as part of an event, it is difficult to detect and usually is not noticed until the fouled sensor is replaced. Fouling most often leads to lower conductivity/salinity readings. For more information on QA related to bio-fouling, see Bushnell et al. (2019).

Advances in TS measurement technology have eliminated many of the problems encountered in older devices. Sensors are smarter, smaller, more reliable, and draw less power. More sensors can be employed and used for comparison to make corrections. Most notably, signal processing hardware and software capabilities have grown substantially. For example, sensor response is more easily digitally characterized and calibrated, as opposed to constructing a physical device with a known response.

2.5 Other Important Considerations

While outside the scope of the real-time tests described in this manual, quality assurance (QA) is critical to data quality. Sensors require attention to proper QA measures both before and after the deployment (Bushnell et al. 2019). Operators must follow the manufacturer's recommendations for factory calibration schedules and proper sensor preparation and maintenance.

Also important, but beyond the scope of this document at present, is the determination and reporting of data uncertainty. Knowledge of the accuracy of each observation is required to ensure that data are used appropriately and aids in the computation of error bounds for subsequent products derived by users. All sensors and measurements contain errors that are determined by hardware quality, calibration accuracy, methods of operation, and data processing techniques. Operators should routinely provide a quantitative measure of data uncertainty in the associated metadata. Such calculations can be challenging, so operators should also document the methods used to compute the uncertainty. The limits and thresholds implemented by operators for the data QC tests described here are a key component in establishing the observational error bounds. Operators are strongly encouraged to consider the impact of the QC tests on data uncertainty, as these two efforts greatly enhance the utility of their data.

Sensor redundancy is key to obtaining reliable measurements and ensuring that uncertainties can be assigned to those measurements. Comparing two adjacent instruments can assist in evaluation of data quality, as well as provide two (or more) independent estimates of a parameter of interest. Variation in the estimates of uncertainty provided by those instruments can occur for several reasons, including water mass gradients in the environment.

3.0 Quality Control

To conduct real-time QC on TS observations, the first pre-requisite is to understand the science and context within which the measurements are being conducted. Each deployment method imposes the need for specific QC methods, with different interpretations of real-time. A fixed TS sensor deployed in coastal waters may report at 5-minute intervals, while deep ocean CTD casts may take hours to provide a profile. While each sensor provides vastly different products, QC techniques can be applied broadly; with the proper selection of thresholds, a check for temporal data spikes in the former is similar to data spike checks in the vertical profile of the latter.

TS measurements can be used to resolve many things, such as internal waves, oceanic fronts, river runoff, upwelling, etc., and some of these can be extreme events. Human involvement is therefore important to ensure that solid scientific principles are applied to data evaluation to ensure that good data are not discarded, and bad data are not distributed.

The real-time QC of TS observations can be extremely challenging. For example, for real-time QC, gradual calibration changes and long-term system responses (sensor drift) most likely cannot be detected or corrected with real-time, automated QC. Drift correction for TS measurements during post-processing is difficult even if a valid post-recovery calibration is obtained. Drift is often caused by bio-fouling, affecting different systems in different ways—a sensor's response will be affected by the added mass of bio-fouling. Another example is the ability of some data providers to backfill data gaps. In both examples, the observations are not considered to be real-time for purposes of QC checks.

3.1 QC Flags

Data are evaluated using QC tests, and the results of those tests are recorded by inserting flags in the data record. Table 3-1 provides a simple set of flags and associated descriptions. Operators may incorporate additional flags for inclusion in metadata records to further assist with troubleshooting. For example, an observation may fail the temperature min/max range test and be flagged as having failed. An operator could provide an additional test to further define a failure: if the data failed the temperature min/max by exceeding the upper limit, a "failed high" flag could indicate that the values were higher than the expected range. Such detailed flags primarily support maintenance efforts and are presently beyond U.S. IOOS requirements for QC of real-time data. For additional information regarding flags, see the *Manual for the Use of Real-Time Oceanographic Data Quality Control Flags* (U.S. IOOS 2017) posted on the U.S. IOOS QARTOD website.

Further post-processing of the data may yield different conclusions from those reached during initial assessments. Flags set in real-time should not be changed to ensure that historical documentation is preserved. Results from post-processing should generate another set of flags.

Observations are time ordered, and the most recent observation is n_0, preceded by a value at n_{-1}, and so on moving back in time. The focus is primarily on the real-time QC of observations n_0, n_{-1}, and n_{-2}.

Table 3-1. Flags for real-time data (UNESCO 2013)

Flag	Description
Pass=1	Data have passed critical real-time quality control tests and are deemed adequate for use as preliminary data.
Not evaluated=2	Data have not been QC-tested, or the information on quality is not available.
Suspect or Of High Interest=3	Data are considered to be either suspect or of high interest to data providers and users. They are flagged suspect to draw further attention to them by operators.
Fail=4	Data are considered to have failed one or more critical real-time QC checks. If they are disseminated at all, it should be readily apparent that they are not of acceptable quality.
Missing data=9	Data are missing; used as a placeholder.

3.2 Test Hierarchy

This section outlines the 13 real-time QC tests that are required, strongly recommended, or suggested for real-time TS measurements. Salinity may be computed onboard the sensor package or after transmission of the raw data. When possible, tests should be applied to conductivity and temperature observations, as well as the derived salinity values, regardless of where the salinity calculation takes place. Operators should also consider that some of these tests can be carried out within the instrument, where thresholds can be defined in configuration files. Although more tests may imply a more robust QC effort, there are many reasons operators could use to justify not conducting some tests. In those cases, operators need only to document reasons these tests do not apply to their observations. Tests are listed in table 3-2 and are divided into three groups: those that are required, strongly recommended, or suggested.

Table 3-2. QC Tests in order of implementation and hierarchy.

Group 1 *Required*	Test 1)	Gap Test
	Test 2)	Syntax Test
	Test 3)	Location Test
	Test 4)	Gross Range Test
	Test 5)	Climatological Test
Group 2 *Strongly Recommended*	Test 6)	Spike Test
	Test 7)	Rate of Change Test
	Test 8)	Flat Line Test
Group 3 *Suggested*	Test 9)	Multi-Variate Test
	Test 10)	Attenuated Signal Test
	Test 11)	Neighbor Test
	Test 12)	TS Curve/Space Test
	Test 13)	Density Inversion Test

3.3 QC Test Descriptions

A variety of tests can be performed to evaluate data quality in real-time. Testing the timely arrival and integrity of the data transmission itself is a first step. If the data are corrupted during transmission, further testing may be irrelevant. The checks defined in these 13 tests evaluate data through various comparisons to other data and to the expected conditions in the given environment. The tests listed in this section presume a time-ordered series of observations and denote the most recent observation as previously described.

Sensor operators need to select the best thresholds for each test, which are determined at the operator level and may require trial and error/iteration before final selections are made. A successful QC effort is highly dependent upon selection of the proper thresholds, which should not be determined arbitrarily but can be based on historical knowledge or statistics derived from more recently acquired data. Although this manual provides some guidance for selecting thresholds based on input from various operators, it is assumed that operators have the expertise and motivation to select the proper thresholds to maximize the value of their QC effort. Operators must openly provide thresholds as metadata for user support. This shared information will help U.S. IOOS document standardized thresholds that will be included in future releases of this manual.

3.3.1 Applications of QC Tests to TS Sensors

These 13 tests require operators to select a variety of thresholds. Examples are provided in the following test tables. However, operators are in the best position to determine the appropriate thresholds for their operations. Some tests rely on multiple data points most recently received to determine the quality of the current data point. When this series of data points reveals that the entire group fails, the current data point is flagged, but the previous flags are not changed. This action supports the view that historical flags are not altered. The first example of this scenario is Test 8, the Flat Line Test.

Test 1) Timing/Gap Test (Required)

Check for arrival of data.		
Test determines that the most recent data point has been measured and received within the expected time window (TIM_INC) and has the correct time stamp (TIM_STMP). **Note:** For those systems that do not update at regular intervals, a large value for TIM_STMP can be assigned. The gap check is not a solution for all timing errors. Data could be measured or received earlier than expected. This test does not address all clock drift/jump issues.		
Flags	**Condition**	**Codable Instructions**
Fail=4	Data have not arrived as expected.	If NOW – TIM_STMP > TIM_INC, flag = 4
Suspect=3	N/A	N/A
Pass=1	Applies for test pass condition.	N/A
Test Exception: None.		
Test specifications to be established locally by the operator. **Example:** TIM_INC = 1 hour		

Test 2) Syntax Test (Required)

Check to ensure that the message is structured properly

Received data message (full message) contains the proper structure without any indicators of flawed transmission such as parity errors. Possible tests are: a) the expected number of characters (NCHAR) for fixed length messages equals the number of characters received (REC_CHAR), or b) passes a standard parity bit check, cyclic redundancy check (CRC), etc. Many such syntax tests exist, and the operator should select the best criteria for one or more syntax tests.

Capabilities for dealing with flawed messages vary among operators; some may have the ability to parse messages to extract data within the flawed message sentence before the flaw. A syntax check is performed only at the message level and not within the message content. In cases where a data record requires multiple messages, this check can be performed at the message level but is not used to check message content.

Flags	Condition	Codable Instructions
Fail=4	Data sentence cannot be parsed to provide a valid observation.	If REC_CHAR ≠ NCHAR, flag = 4
Suspect =3	N/A	N/A
Pass=1	Expected data sentence received; absence of parity errors.	

Test Exception: None.

Test specifications to be established locally by the operator.
Example: NCHAR = 128

Test 3) Location Test (Required)

Check for reasonable geographic location.

Test checks that the reported present physical location (latitude/longitude) is within operator-determined limits. The location test(s) can vary from a simple impossible location to a more complex check for displacement (DISP) exceeding a distance limit (RANGEMAX) based upon a previous location and platform speed. Operators may also check for erroneous locations based upon other criteria, such as reported positions over land, as appropriate.

Flags	Condition	Codable Instructions
Fail=4	Impossible location.	|LAT | > 90 or |LONG | > 180
Suspect=3	Unlikely platform displacement.	DISP > RANGEMAX
Pass=1	Applies for test pass condition.	N/A

Test Exception: Test does not apply to fixed deployments when no location is transmitted.

Test specifications to be established locally by the operator.
Example: Displacement DISP calculated between sequential position reports, RANGEMAX = 20 km

Test 4) Gross Range Test (Required)

Data point exceeds sensor or operator-selected min/max. Applies to T, SP, C and P.

All sensors have a limited output range, and this can form the most rudimentary gross range check. No values less than a minimum value or greater than the maximum value the sensor can output (T_SENSOR_MIN, T_SENSOR_MAX) are acceptable. Additionally, the operator can select a smaller span (T_USER_MIN, T_USER_MAX) based upon local knowledge or a desire to draw attention to extreme values.

NOTE: Operators may choose to flag as suspect values that exceed the calibration span but not the hardware limits (e.g., a value that sensor is not capable of producing or negative conductivity).

Flags	Condition	Codable Instructions
Fail=4	Reported value is outside of sensor span.	If T_n < T_SENSOR_MIN, or T_n > T_SENSOR_MAX, flag = 4
Suspect=3	Reported value is outside of operator-selected span.	If T_n < T_USER_MIN, or T_n > T_USER_MAX, flag = 3
Pass=1	Applies for test pass condition.	

Test Exception: None.

Test specifications to be established locally by the operator.

Examples: The following global range min/max are applied on some climate and forecast standard-names in the IMOS toolbox: depth: -5/12,000 m
sea_water_pressure: -5/12,000 decibars (dbar)
sea_water_pressure_due_to_sea_water: -15/12,000 dbar
sea_water_salinity: 2/41
sea_water_temperature: -2.5/40 °C

Test 5) Climatology Test (Required)

Test that data point falls within seasonal expectations. Applies to T and SP.

This test is a variation on the gross range check, where the thresholds T_Season_MAX and T_Season_MIN are adjusted monthly, seasonally, or at some other operator-selected time period (TIM_TST). Expertise of the operator is required to determine reasonable seasonal averages. Longer time series permit more refined identification of appropriate thresholds. The ranges should also vary with water depth, if the measurements are taken at sites that cover significant vertical extent and if climatological ranges are meaningfully different at different depths (e.g., narrower ranges at greater depth).

Flags	Condition	Codable Instructions
Fail=4	Because of the dynamic nature of T and S in some locations, no fail flag is identified for this test.	N/A
Suspect=3	Reported value is outside of operator-identified climatology window.	If T_n < T_Season_MIN or T_n > T_Season_MAX, flag = 3
Pass=1	Applies for test pass condition.	N/A

Test Exception: None.

Test specifications to be established locally by operator: A seasonal matrix of T_{max} and T_{min} values at all TIM_TST intervals**.**

Examples: T_SPRING_MIN = 12 °C, T_SPRING_MAX = 18.0 °C

Test 6) Spike Test (Strongly Recommended)

Data point *n*-1 exceeds a selected threshold relative to adjacent data points. Applies to T, SP, C, and P.

This check is for single value spikes, specifically the value at point *n*-1. Spikes consisting of more than one data point are difficult to capture, but their onset may be flagged by the rate of change test. The spike test consists of two operator-selected thresholds, THRSHLD_LOW and THRSHLD_HIGH. Adjacent data points (n_{-2} and n_0) are averaged to form a spike reference (SPK_REF). The absolute value of the spike is tested to capture positive and negative spikes. Large spikes are easier to identify as outliers and flag as failures. Smaller spikes may be real and are only flagged suspect. The thresholds may be fixed values or dynamically established (for example, a multiple of the standard deviation over an operator-selected period).

Flags	Condition	Codable Instructions
Fail=4	High spike threshold exceeded.	If $\lvert T_{n-1} - SPK_REF \rvert >$ THRSHLD_HIGH, flag = 4
Suspect=3	Low spike threshold exceeded.	If $\lvert T_{n-1} - SPK_REF \rvert >$ THRSHLD_LOW and $\lvert T_{n-1} - SPK_REF \rvert \leq$ THRSHLD_HIGH, flag = 3
Pass=1	Applies for test pass condition.	N/A

Test Exception: None.

Test specifications to be established locally by the operator.
Examples: THRSHLD_LOW = 3 °C, THRSHLD_HIGH = 8 °C

Test 7) Rate of Change Test (Strongly Recommended)

Excessive rise/fall test. Applies to T, SP, C, and P.

This test inspects the time series for a time rate of change that exceeds a threshold value identified by the operator. T, SP, C, P values can change substantially over short periods in some locations, hindering the value of this test. A balance must be found between a threshold set too low, which triggers too many false alarms, and one set too high, making the test ineffective. Determining the excessive rate of change is left to the local operator.

The following shows two different examples of ways to select the thresholds provided by QARTOD VI participants. Implementation of this test can be challenging. Upon failure, it is unknown which of the points is bad. Further, upon failing a data point, it remains to be determined how the next iteration can be handled.

Example 1

The rate of change between temperature T_{n-1} and T_n must be less than three standard deviations (3*SD). The SD of the T time series is computed over the previous 25-hour period (operator-selected value) to accommodate cyclical diurnal and tidal fluctuations. Both the number of SDs (N_DEV) and the period over which the SDs (TIM_DEV) are calculated and determined by the local operator.

Example 2

The rate of change between temperature T_{n-1} and T_n must be less than 2 °C +2SD.

$|T_{n-1} - T_{n-2}| + |T_{n-1} - T_n| <= 2*N_DEV*SD$ (example provided by EuroGOOS).

Flags	Condition	Codable Instructions
Fail=4	No fail flag is identified for this test.	N/A
Suspect=3	The rate of change exceeds the selected threshold.	If $\lvert T_n - T_{n-1} \rvert >$ N_DEV*SD, flag = 3
Pass=1	Applies for test pass condition.	N/A

Test Exception: None.

Test specifications to be established locally by operator.
Example: N_DEV = 3, TIM_DEV = 25

Test 8) Flat Line Test (Strongly Recommended)

Invariant value. Applies to T, SP, C, and P.

When some sensors and/or data collection platforms fail, the result can be a continuously repeated observation of the same value. This test compares the present observation *n* to a number (REP_CNT_FAIL or REP_CNT_SUSPECT) of previous observations. Observation *n* is flagged if it has the same value as previous observations within a tolerance value, EPS, to allow for numerical round-off error. Note that historical flags are not changed.

Flags	Condition	Codable Instructions
Fail=4	When the five most recent observations are equal, T_n is flagged fail.	CNT = 0 For *I* = 1,REP_CNT_FAIL If $\|T_n - T_{n-i}\|$<EPS, CNT = CNT+1 If CNT = REP_CNT_FAIL, flag = 4
Suspect=3	It is possible but unlikely that the present observation and the two previous observations would be equal. When the three most recent observations are equal, T_n is flagged suspect.	CNT = 0 For *I* = 1,REP_CNT_SUSPECT If $\|T_n - T_{n-i}\|$<EPS, CNT = CNT+1 If CNT = REP_CNT_SUSPECT, flag = 3
Pass=1	Applies for test pass condition.	N/A

Test Exception: None.

Test specifications to be established locally by the operator.
Examples: REP_CNT_FAIL = 5, REP_CNT_SUSPECT= 3, EPS = 0.05°

Test 9) Multi-Variate Test (Suggested)

Comparison to other variables. Applies to T, SP, and P.

This is an advanced family of tests, starting with the simpler test described here and anticipating growth towards full co-variance testing in the future. It is doubtful that anyone is conducting tests such as these in real time. As these tests are developed and implemented, they should be documented and standardized in later versions of this manual.

This example pairs rate of change tests as described in test 7. The T (or SP or P) rate of change test is conducted with a more restrictive threshold (N_T_DEV). If this test fails, a second rate of change test operating on a second variable (salinity or conductivity would be the most probable) is conducted. The absolute value rate of change should be tested, since the relationship between T and variable two is indeterminate. If the rate of change test on the second variable fails to exceed a threshold (e.g., an anomalous step is found in T and is lacking in salinity), then the T_n value is flagged.

Note that Test 12, TS Curve/Space Test is a well-known example of the multi-variate test.

Flags	Condition	Codable Instructions
Fail=4	No fail flag is identified for this test.	N/A
Suspect=3	T_n fails the rate of change and the second variable does not exceed the rate of change.	If $\lvert T_n - T_{n-1}\rvert$>N_T_DEV*SD_T AND $\lvert SP_n - SP_{n-1}\rvert$<N_SP_DEV*SD_SP, flag = 3
Pass=1	N/A	N/A

Test Exception: None.

Test specifications to be established locally by the operator.
Examples: N_T_DEV = 2, N_TEMP_DEV = 2, TIM_DEV = 25 hours

In a more complex case, more than one secondary rate of change test can be conducted. Temperature, salinity, turbidity, nutrients, and chlorophyll are all possible secondary candidates, and all could be checked for anomalous rate of change values. In this case, a knowledgeable operator may elect to pass a high rate of change observation when any one of the secondary variables also exhibits a high rate of change. Such tests border on modeling, should be carefully considered, and may be beyond the scope of this effort.

The QARTOD TS committee recognized the high value in full co-variance testing but also noted the challenges. Therefore full co-variance QC tests are still considered experimental.

Test 10) Attenuated Signal Test (Suggested)

A test for inadequate variation of the time series. Applies to T, SP, C, and P.

A common sensor failure mode can provide a data series that is nearly but not exactly a flat line (e.g., if the sensor head were to become wrapped in debris). This test inspects for an SD value or a range variation (MAX-MIN) value that fails to exceed threshold values (MIN_VAR_WARN, MIN_VAR_FAIL) over a selected time period (TST_TIM).

Flags	Condition	Codable Instructions
Fail=4	Variation fails to meet the minimum threshold MIN_VAR_FAIL.	If During TST_TIM, SD <MIN_VAR_FAIL, or During TST_TIM, MAX-MIN <MIN_VAR_FAIL, flag = 4
Suspect=3	Variation fails to meet the minimum threshold MIN_VAR_WARN.	If During TST_TIM, SD <MIN_VAR_WARN, or During TST_TIM, MAX-MIN <MIN_VAR_WARN, flag = 3
Pass=1	Applies for test pass condition.	N/A

Test Exception: None.

Test specifications to be established locally by the operator.
Examples: TST_TIM = 12 hours
MIN_VAR_WARN = 0.5 °C, MIN_VAR_FAIL = 0.1 °C

Test 11) Neighbor Test (Suggested)

Comparison to nearby sensors. Applies to T, SP, C, and P.

This test is potentially the most useful when a nearby sensor has a similar response. Ideally, redundant sensors using different technology would be co-located and alternately serviced at different intervals. This close neighbor would provide the ultimate QC check, but cost often prohibits such a deployment

However, there are few instances where a second sensor is sufficiently proximate to provide a useful QC check. Just a few hundred meters in the horizontal and less than 10 m vertical separation can often yield greatly different results. Nevertheless, the test should not be overlooked where it may have application.

This test is the same as Test 9), *Multi-variate Check – comparison to other variables* where the second variable is the second sensor. The selected thresholds depend entirely upon the relationship between the two sensors as determined by the local knowledge of the operator.

In the instructions and examples below, data from one site (T1) are compared to a second site (T2). The standard deviation for each site (SD1, SD2) is calculated over the period (TIM_DEV) and multiplied as appropriate (N_T1_DEV for site T1) to calculate the rate of change threshold. Note that an operator could also choose to use the same threshold for each site, since they are presumed to be similar.

Flags	Condition	Codable Instructions
Fail=4	No fail flag is identified for this test.	N/A
Suspect=3	$T1_n$ fails the rate of change and the second sensor $T2_n$ does not exceed the rate of change.	If $\lvert T1_n - T1_{n-1}\rvert$>N_T1_DEV*SD1 AND $\lvert T2_n - T2_{n-1}\rvert$<N_T2_DEV*SD2, flag = 3
Pass=1	N/A	N/A

Test Exception: There is no adequate neighbor.

Test specifications to be established locally by the operator.
Examples: N_T1_DEV = 2, N_T2_DEV=2, TIM_DEV = 25 hours

Test 12) TS Curve/Space Test (Suggested)

Comparison to expected TS relationship. Applies to T, SP.

The TS curve is a classic tool used to evaluate observations, especially in the open ocean below the thermocline. Site-specific TS curve characteristics are used to identify outliers. The curve could be either a fitted equation or numerical table. For a given T_n, SP_n is expected to be within SPfit $\pm$ SP_fit_warn or SP_fit_fail, operator-provided values. The value SPfit is obtained from the equation or table.

Flags	Condition	Codable Instructions
Fail=4	For a given temperature, the observed salinity falls outside the TS curve failure threshold.	If $\lvert SP_n\text{-SPfit}\rvert >$ SP_fit_fail, flag = 4
Suspect=3	For a given temperature, the observed salinity falls outside the TS curve warning threshold.	If $\lvert SP_n\text{-SPfit}\rvert \leq$ SP_fit_fail and $\lvert SP_n\text{-SPfit}\rvert >$ SP_fit warn, flag = 3
Pass=1	N/A	N/A

Test Exception: The test will probably not be useful in estuaries or ocean surface waters.

Test specifications to be established locally by the operator.

Examples: At the Bermuda Atlantic Time Series site, for a temperature of 18 °C, SPfit = 36.5
SP_fit_fail = 0.05, SP_fit_warn = 0.02

Test 13) Density Inversion Test (Suggested)

Checks that density increases with pressure (depth).

With few exceptions, potential water density σ_θ will increase with increasing pressure. When vertical profile data are obtained, this test is used to flag as failed T, C, and SP observations, which yield densities that do not sufficiently increase with pressure. A small, operator-selected density threshold (DT) allows for micro-turbulent exceptions. Here, $\sigma_{\theta n}$ is defined as one sample increment deeper than $\sigma_{\theta n-1}$. With proper consideration, the test can be run on downcasts, upcasts, or down/up cast results produced in real-time.

From a computational point of view, this test is similar to the rate of change test (test 7), except that the time axis is replaced by depth. The same code can be used for both, using different variables and thresholds. As with the rate of change test, it is not known which side of the step is good versus bad.

An example of the software to compute sigma-theta is available at http://www.teos-10.org/software.htm.

Flags	Condition	Codable Instructions
Fail=4	Potential density does not sufficiently increase with increasing depth.	If $\sigma_{\theta n-1}$+DT > $\sigma_{\theta n}$, flag = 4
Suspect=3	No suspect flag is identified for this test.	N/A
Pass=1	Potential density sufficiently increases with increasing depth.	If $\sigma_{\theta n-1}$+DT $\leq \sigma_{\theta n}$, flag = 1

Test Exception: None.

Test specifications to be established locally by the operator.

Examples: DT = 0.03 kg/m^3

3.3.2 Applications of QC Tests to Mobile TS Sensors

The specific application of the QC tests can be dependent on the way the sensor is deployed. For mobile platforms, at least two existing programs, GTSPP and Argo, have developed QC tests that are similar to the U.S. IOOS QARTOD tests in this manual. Manuals from both programs are available online (IOC 2010; Argo 2019). Operators within such programs will likely find their present QC process to be compliant with U.S. IOOS QARTOD requirements and recommendations, which is the intention of the QARTOD TS Committee. Table 3-3 provides a comparison of salinity and temperature QC tests from the U.S. IOOS QARTOD, GTSPP, and real-time Argo programs.

Table 3-3. Comparison of QARTOD, GTSPP, and Argo temperature and salinity QC tests

QARTOD	GTSPP Manual[1]	Argo Manual[2]
1) Time/Gap Test	1.2	2
2) Syntax Test	No match	1 (close, not identical)
3) Location Test	1.3, 1.4	3, 4, 5
4) Gross Range Test	2.1	6, 7
5) Climatological Test	3.1, 3.2, 3.3, 3.4	No match
6) Spike Test	2.7, 2.8	9
7) Rate of Change Test	2.9, 4.1	11
8) Flat Line Test	2.4, 2.5	14, 18
9) Multi-Variate Test	No match	No match
10) Attenuated Signal Test	2.4	16 (close, not identical)
11) Neighbor Test	No match	No match
12) TS Curve/Space Test	No match	No match
13) Density Inversion Test	2.10	14

Tables 3-4 through 3-6 provide a summary of each QC test described in section 3.3.1 and indicate any changes necessary for the test to be applied to mobile deployment scenarios. Note that the "s" axis indicates "along path" for mobile platforms. Each data point, whether horizontal, vertical, or along the path, is quality controlled and assigned a flag using these tests. Operators may choose to expand upon the flagging scheme using another tier of flags, e.g., to characterize the entire vertical profile.

[1] IOC 2010

[2] Wong et al. 2020

Table 3-4. Application of Required QC Tests (Tests 1-5) for TS Sensor Deployments

Test	Condition	Platform	Codable Instructions
1) Timing/Gap Test (Required)	Check for arrival of data.	Stationary	No change
		Fixed Vertical	
		Mobile	
		3D	
2) Syntax Test (Required)	Expected data sentence received, absence of parity errors.	Stationary	No change
		Fixed Vertical	
		Mobile	
		3D	
3) Location Test (Required)	Check for acceptable geographic location.	Stationery	No change
		Fixed Vertical	
		Mobile	
		3D	
4) Gross Range Test (Required)	Data point exceeds sensor or operator selected min/max.	Stationary	No change
		Fixed Vertical	
		Mobile	
		3D	
5) Climatology Test (Required.	Test that data point falls within seasonal expectations.	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	Test is conducted along s, x, or y axis.
		3D	Test is conducted along s, x, y, or z axis.

Table 3-5. Application of Strongly Recommended QC Tests (Tests 6-8) for TS Sensor Deployments

Test	Condition	Platform	Codable Instructions
6) Spike Test (Strongly Recommended)	Data point *n*-1 exceeds a selected threshold relative to adjacent data points.	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	No change, or test is conducted along s, x, or y axis.
		3D	No change, or test is conducted along s, x, y, or z axis.
7) Rate of Change Test (Strongly Recommended)	Excessive rise/fall test.	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	No change, or test is conducted along s, x, or y axis.
		3D	No change, or test is conducted along s, x, y, or z axis.
8) Flat Line Test (Strongly Recommended)	Invariant value	Stationary	No change
		Vertical	Test is conducted along z axis.
		Mobile	No change, or test is conducted along s, x, or y axis.
		3D	No change, or test is conducted along s, x, y, or z axis.

Table 3-6. Application of Suggested QC Tests (Tests 9-13) for TS Sensor Deployments

Test	Condition	Platform	Codable Instructions
9) Multi-Variate Test (Suggested)	Comparison to other variables	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	Test is conducted along s, x, or y axis.
		3D	Test is conducted along s, x, y, or z axis.
10) Attenuated Signal Test (Suggested)	Inadequate variation test	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	No change, or test is conducted along s, x, or y axis.
		3D	No change, or test is conducted along s, x, y, or z axis.
11) Neighbor Test (Suggested).	Comparison to nearby sensors of the same variable	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	No change
		3D	No change
Test 12) TS Curve/Space Test (Suggested)	Comparison to expected TS relationship	Stationary	No change
		Fixed Vertical	Test is conducted along z axis.
		Mobile	Test is conducted along s, x, or y axis.
		3D	Test is conducted along s, x, y, or z axis.
Test 13) Density Inversion Test (Suggested)	Checks that density increases with pressure (depth)	Stationary	N/A
		Fixed Vertical	Test is conducted along z axis.
		Mobile	Test is conducted along z axis.
		3D	Test is conducted along z axis.

4.0 Summary

The QC tests in this TS manual have been compiled using the guidance provided by the TS committee and valuable reviewers (appendix A), all QARTOD workshops (QARTOD 2003-2009) and earlier U.S. IOOS QARTOD manuals. Test suggestions came from both operators and TS data users with extensive experience. The considerations of operators who ensure the quality of real-time data may be different from those whose data are not published in real-time, and these and other differences must be balanced according to the specific circumstances of each operator. Although these real-time tests are required, strongly recommended, or suggested, it is the operator who is responsible for deciding which tests are appropriate. Each operator selects thresholds based on the specific program requirements that must be met. The scope of requirements can vary widely—from complex data streams that support myriad QC checks to ensure precise and accurate measurements to basic data streams that do not need such details. Operators must publish their QC processes via metadata so that data users can readily see and understand the source and quality of those data.

The 13 QC tests identified in this manual apply to TS observations from a variety of sensor types and platforms that may be used in U.S. IOOS. At least two existing programs, GTSPP (IOC 2010) and Argo (Argo 2019), have developed QC tests for mobile platforms that are similar to the U.S. IOOS QARTOD tests in this manual. The QARTOD TS committee intends for the QC tests of these programs to be compliant with U.S. IOOS QARTOD requirements and recommendations. The individual tests are described and include codable instructions, output conditions, example thresholds, and exceptions (if any).

Selection of the proper thresholds is critical to a successful QC effort. Thresholds can be based on historical knowledge or statistics derived from more recently acquired data, but they should not be determined arbitrarily. This manual provides guidance for selecting thresholds based on input from various operators, but also notes that operators need the subject matter expertise and motivation to select the proper thresholds to maximize the value of their QC effort.

Future QARTOD reports will address standard QC test procedures and best practices for all types of common and uncommon platforms and sensors for all the U.S. IOOS core variables. Some test procedures may take place within the sensor package. Significant components of metadata will reside in the sensor and be transmitted either on demand or automatically along with the data stream. Users may also reference metadata through Uniform Resource Locators (URLs) to simplify the identification of which QC steps have been applied to data. However, QARTOD QC test procedures in this manual address only real-time in-situ observations. The tests do not include post-processing, which is not in real-time but may be useful for ecosystem-based management, or delayed-mode, which might be suitable for climate studies.

Each QARTOD manual is envisioned as a dynamic document and will be posted on the QARTOD website at https://ioos.noaa.gov/project/qartod/. This process allows for QC manual updates as technology development occurs for both upgrades of existing sensors and new sensors.

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Ocean Data Standards Resources
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